EPOCH issue and improvement register

Contributors to this register

Official `main` status checked 30 October 2026

1 Purpose and status convention

This is a shared register of EPOCH bugs and incremental improvements. It starts with the official EPOCH source and records whether each change is in official `main`. An implementation in a separate fork is identified explicitly; it does not by itself make an upstream issue fixed. The register is selective, and contributors are welcome to add independently verified issues from any part of EPOCH.

The official `main` tip checked here is `f294c484f76dff0777d5cc0d50b38506a2b049ff`. The statuses mean: **Fixed**—the correction is in that `main` history; **Dev only**—it is in an official development branch but not `main`; **Open**—the stated stock-code behaviour remains in `main`; **Baseline**—an existing method works as designed but has a documented alternative; **N/A**—the affected feature exists only in the fork, so this is not a stock EPOCH bug. Statuses are source assessments at the stated commit, not fresh simulation results.

2 Registry

The linked ID opens the detailed entry. E IDs describe stock EPOCH; F IDs mark fork-only features. To add an item, copy one `\RegistryRow` line in `registry.tex` and add a matching labelled entry in `items.tex`; the title column wraps automatically.

ID	Short description	Main status
E01	Laser absorption diagnostic at maximum boundaries	Fixed
E02	2D linear Breit–Wheeler species selection	Fixed
E03	Photon probes can deadlock with particle IDs	Dev only
E04	Particle IDs can be reused after restart or file loading	Open
E05	Higuera–Cary magnetic rotation uses the wrong field scale	Open
E06	Trident can produce pairs when pair production is disabled	Open
E07	NBW pair generation uses the wrong photon species index	Open
E08	`identify` silently replaces explicit charge or mass	Open
E09	NBW uses approximate rate and energy-split tables	Baseline
F01	Restore spin state after restart	N/A
F02	Correct boosted pair-channel sampling and scaling	N/A
F03	Polarisation-resolved NBW energy-split tables	N/A
F04	Spin-resolved NLCS quantile sampling	N/A
F05	Sub-step spin and polarisation no-transition update	N/A
F06	Full Landau–Lifshitz force and midpoint RR kick	N/A
F07	Custom-laser half-cell sampling offset	N/A
F08	Custom-laser MPI file-loading deadlock	N/A
F09	Independent custom-file storage for each laser	N/A
F10	Continuous custom-profile re-slabbing in 3D	N/A

3 Stock EPOCH: fixes already in `main`

3.1 E01: Laser absorption diagnostic at maximum boundaries

Failure. Energy injected at `x_max`, `y_max` or `z_max` was recorded with the wrong sign, which could yield a zero absorption coefficient. The change touches `laser.f90` in all three dimensions.

Upstream status. Tom Goffrey merged PR #896 into `4.20-devel` on 10 April 2026 as `0984b1d31`; that merge is an ancestor of the checked official `main` tip. The v4.20.1 release notes list this fix (issue #893). The present register does not repeat the upstream test suite.

3.2 E02: 2D linear Breit–Wheeler species selection

Failure and status. A typo in the 2D linear Breit–Wheeler routine selected the wrong species. PR #892 merged `ba04ae645` into `4.20-devel` on 26 March 2026; it is an ancestor of official `main` and is listed in v4.20.1 (issue #891). This is a separate process from the nonlinear Breit–Wheeler issues below.

4 Stock EPOCH: development-only or open items

4.1 E03: Photon-probe particle-ID deadlock

4.1.1 Trigger and failure

With photon probes and `PARTICLE_ID` or `PARTICLE_ID4`, a rank with no local photons could return before the collective ID-generation call. Other ranks entered it and waited indefinitely.

4.1.2 Official and fork status

PR #906 merged `241d6fb9c` into official `4.20-devel` on 1 June 2026, in all three dimensions. That commit is not an ancestor of the checked `main` tip. The later 2D fork commit `3e4734522` also fixes the photon path and includes the distinct ID-lifetime work in E04. Do not treat the two commits as equivalent in scope.

4.1.3 Verification to record

A regression test should use at least two MPI ranks, a photon probe and a rank without local photons; the run must complete and IDs must be unique. A successful single-rank run cannot test this collective failure.

4.2 E04: Particle-ID reuse across restart and file loading

Failure. In the 2D fork investigation, the maximum assigned ID was not reliably carried through restart and particles loaded from file; newly created particles could reuse existing IDs. The main source at the checked tip does not contain the counter persistence and raising logic of `3e4734522`. This is separate from the photon-probe deadlock in E03.

Evidence and limit. The fork commit changes `epoch2d/src/housekeeping/setup.F90`, `partlist.F90` and related paths. A portable reproducer should check uniqueness after particles leave the domain, after restart with a changed MPI decomposition, and after both automatic and supplied file IDs. The registered correction is 2D only.

4.3 E05: Higuera–Cary field normalisation

4.3.1 Failure and correction

The stock Higuera–Cary magnetic rotation forms its coefficient from `part_q * dt / part_m` and the interpolated magnetic field. The interpolation weights need the same normalisation used by the other particle push; without it, the rotation uses the wrong field scale. The fork fix across all three dimensions `d86ceceab` instead uses `ccmratio`, which includes that factor. An isolated variant based on official `4.21-devel` is `2684956bf`.

4.3.2 Official status and test

The corrected expression is absent from the checked official `main` source. A suitable regression is a crossed-field single-particle push with Higuera–Cary enabled, compared against a known analytic or independently integrated trajectory across particle shape choices. The source review here does not replace that runtime test on an official build.

4.4 E06: Trident event despite `produce_pairs = F`

Trigger and failure. With `TRIDENT_PHOTONS` compiled and pair production disabled in the deck, the electron loop could still decrement trident optical depth and create a pair. If no trident pair species had been designated, the event could access species index `-1` and crash.

Correction and status. `1a88d4afa` guards both the depth decrement and event creation with `produce_pairs` in 1D, 2D and 3D. The checked official `main` still contains the unguarded trident path. A regression should compile the trident define, run with `produce_pairs = F` until a would-be event, and check that no pair is created and the run completes.

4.5 E07: Wrong photon-species argument in nonlinear BW

Failure. In the stock 2D photon loop, pair generation receives the global `photon_species` rather than the current `ispecies`. With multiple photon species this can associate a pair event with the wrong list and may crash. The 2D fork change `2b18ef6b4` passes `ispecies`; the checked official `main` retains the former argument.

Verification to record. Use at least two photon species that can undergo nonlinear Breit–Wheeler conversion. Check species accounting and that both event paths finish without a list error.

4.6 E08: `identify` can replace explicit charge and mass

Behaviour. The stock species parser can apply `identify` after explicit `charge` and `mass` values, silently replacing them. This may make a deck simulate a different particle species than its author intended. The all-dimension fork commit `b908c8591` adds a warning for mismatched explicit values; it does not change the selected physics. The warning is absent from the checked official `main`. A regression should cover both input orders, matching and mismatching values, and each dimension.

4.7 E09: Approximate nonlinear BW tables

Baseline. Official 2D EPOCH reads `pairprod.table`, whose nonlinear Breit–Wheeler rate uses Erber's approximate analytic expression. The existing `energy_split.table` is likewise not a general spin-resolved LCFA spectrum. These are established model choices, so this entry is labelled **Baseline**, not a software crash or a demonstrated upstream regression.

Fork alternative. The 2D fork adds the numerically tabulated LCFA rate `pairprod_lcfa.table` in `848d21a4b` and the `pair_tables = legacy | lcfa` selector in `fd56a2e08`. With spin QED off, `legacy` is the default; with spin QED on, `lcfa` is the default. When spin is on, `legacy` restores the old rate only; polarisation-resolved energy splitting remains. The generator scripts are currently untracked in the local fork, so clean-checkout table regeneration remains to be documented before claiming a reproducible upstream proposal. Runs should record the selector and table hashes.

5 Fork-only features: separate from stock EPOCH defects

The F entries are changes to features absent from the checked official `main`. They are retained here as potentially useful upstream work, and are deliberately marked **N/A** rather than *open bugs* in stock EPOCH.

5.1 Spin and pair channels

F01: Spin restart. The 2D spin branch restores stored spin state during restart (`a66b394cb`). Stock `main` has no corresponding spin-aware restart feature in this comparison.

F02: Boosted pair channels. The fork's pair-boost work includes Breit–Wheeler boost handling (`3e9f3121f`), muon gamma sampling (`fbb9a9459`), Bethe–Heitler scaling (`92552de69`, `a5716caf2`) and additive optical depths (`99ce3c685`). These are distinct patches to the fork's boosted pair channels, not one stock-code fix.

F03: Polarised NBW energy split. The 2D spin implementation adds parallel/perpendicular nonlinear Breit–Wheeler energy-split tables and final-state sampling (`dfb93b141`, `2b18ef6b4`). This is separate from the unpolarised rate alternative in E09.

F04: NLCS quantile sampling. The 2D spin implementation adds inverse-CDF tables for spin-resolved nonlinear Compton photon energies, selected by `use_nlcs_quantile_tables` (`6610b3768`). Spin-off runs continue to use the legacy `ksi_sokolov.table` path.

F05: No-transition update. The 2D spin implementation adds a sub-step spin/polarisation no-transition update (`944fff881`); it is not a global change to every EPOCH QED mode.

5.2 Radiation reaction

F06: Full Landau–Lifshitz and midpoint kick. The 2D fork adds `use_full_landau_lifshitz`, an alternative `gaunt_factor` fit and `rr_midpoint` (`dfa48930a`). The defaults retain the fork's old behaviour. The audited branch has no equivalent 3D implementation.

5.3 Custom-file laser injection

The official `main` tip has no custom-file laser reader. The upstream custom-profile issue #910 is an issue, not an upstream merge. Consequently F07–F10 describe corrections to the fork feature.

F07: Half-cell sampling. The 2D custom reader corrects a half-cell coordinate offset (`1b307b37d`).

5.3.1 F08: MPI file-loading deadlock

The 2D file-profile and phase-loading path could enter MPI operations inconsistently across ranks. `5a77c367b` makes the load collective-safe. Validation should include multiple ranks, profile and phase files, and a rank whose local slab has no requested data.

F09: Per-laser file storage. The 2D reader stores imported profiles with each laser block, so two distinct lasers do not alias the same data (`945fd060d`).

F10: Continuous 3D re-slabbing. The 3D reader reloads the spatiotemporal profile window as load balancing moves the domain slab (`f572d035c`).

6 Adding and updating entries

Use the short table row for discovery, and keep reproduction steps, source paths, links, affected dimensions and validation in the corresponding detailed item. For a serious defect, use a `\subsection` and, if useful, `\subsubsection` headings. Give fork-specific additions an F ID and state which feature is absent from stock EPOCH. Do not mark an item **Fixed** merely because a PR was merged to a development branch. Record the official `main` commit and check date whenever a status changes. A copyable example and the build instructions are in `CONTRIBUTING.md`.